

Delocalization and conjugation

Connections

➡ Building on

- Orbitals and bonding [ch4](#)
- Representing mechanisms by curly arrows [ch5](#)
- Ascertaining molecular structure spectroscopically [ch3](#)

Arriving at

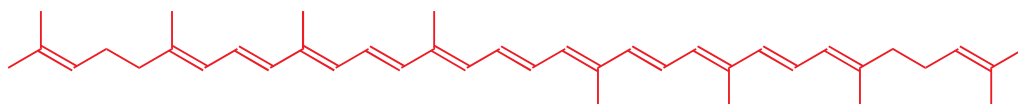
- Interaction between orbitals over many bonds
- Stabilization by the sharing of electrons over more than two atoms
- Where colour comes from
- Molecular shape and structure determine reactivity
- Representing one aspect of structure by curly arrows
- Structure of aromatic compounds

➡ Looking forward to

- Acidity and basicity [ch8](#)
- How conjugation affects reactivity [ch10](#), [ch11](#), & [ch15](#)
- Conjugate addition and substitution [ch22](#)
- Chemistry of aromatic compounds [ch21](#) & [ch22](#)
- Enols and enolates [ch20](#), [ch24–ch27](#)
- Chemistry of heterocycles [ch29](#) & [ch30](#)
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- Chemistry of life [ch42](#)

Introduction

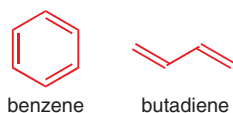
As you look around you, you will be aware of many different colours—from the greens and browns outside to the bright blues and reds of the clothes you are wearing. All these colours result from the interaction of light with the pigments in these different things—some frequencies of light are absorbed, others scattered. Inside our eyes, chemical reactions detect these different frequencies and convert them into electrical nerve impulses sent to the brain. All these pigments have one thing in common—lots of double bonds. For example, the pigment responsible for the red colour in tomatoes, lycopene, is a long-chain polyalkene.



lycopene, the red pigment in tomatoes, rose hips, and other berries

Lycopene contains only carbon and hydrogen; many pigments contain other elements. But nearly all contain double bonds—and many of them. This chapter is about the properties, including colour, of molecules that have several double bonds. These properties depend on the way the double bonds join up, or *conjugate*, and the resulting *delocalization* of the electrons within them.

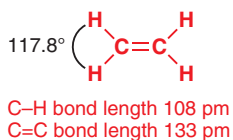
In earlier chapters we talked about carbon skeletons made up of σ bonds. In this chapter we shall see how, in some cases, we can also have a large π framework spread over many atoms and how this dominates the chemistry of such compounds. We shall see how this π framework is responsible for the otherwise unexpected *stability* of certain cyclic polyunsaturated



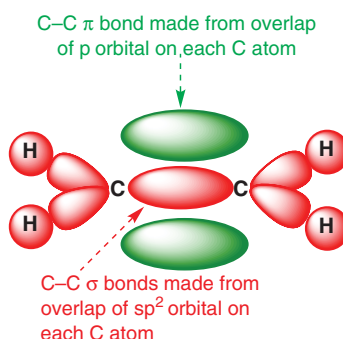
compounds, including benzene, but also *reactivity* in others, such as butadiene. We shall also see how this π framework gives rise to colour. To understand such molecules properly, we need to start with the simplest of all unsaturated compounds, ethene.

The structure of ethene (ethylene, $\text{CH}_2=\text{CH}_2$)

The structure of ethene (ethylene) is well known. It has been determined by electron diffraction and is **planar** (all atoms are in the same plane), with the bond lengths and angles shown on the left. The carbon atoms are roughly trigonal and the $\text{C}=\text{C}$ bond distance is shorter than that of a typical $\text{C}-\text{C}$ single bond. The electronic structure of ethene, you will recall from Chapter 4, can be considered in terms of two sp^2 hybridized C atoms with a σ bond between them and four σ bonds linking them each to two H atoms. The π bond is formed by overlap of a p orbital on each carbon atom.

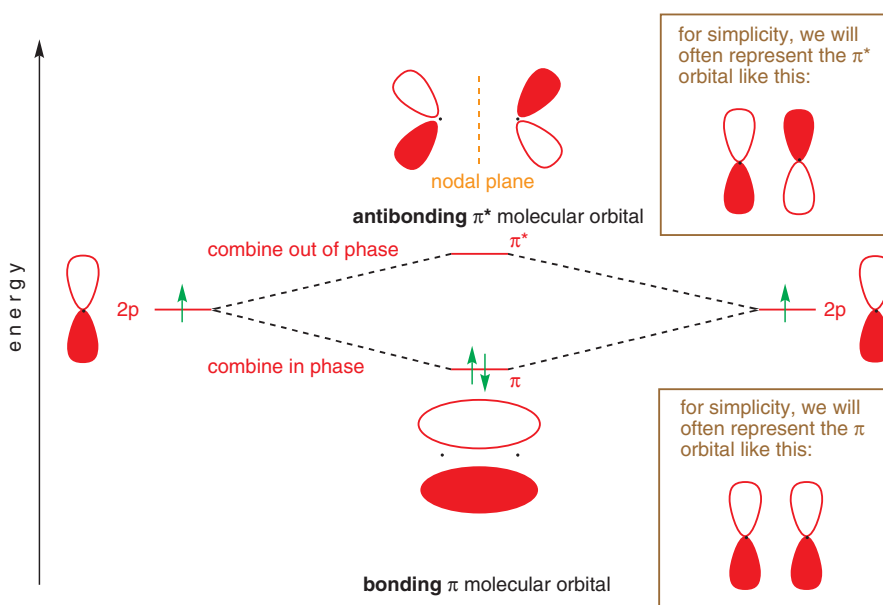


Interactive bonding orbitals in ethene



Ethene is chemically more interesting than *ethane* because of the π system. As you saw in Chapter 5, alkenes can be nucleophiles because the electrons in the π bond are available for donation to an electrophile. But remember that when we combine *two* atomic orbitals we get *two* molecular orbitals, from combining the p orbitals either in phase or out of phase. The in-phase combination accounts for the bonding molecular orbital (π), whilst the out-of-phase combination accounts for the antibonding molecular orbital (π^*). The shapes of the orbitals as they were introduced in Chapter 4 are shown below, but in this chapter we will also represent them in the form shown in the brown boxes—as the constituent p orbitals.

➡ We described the structure of ethene in Chapter 4 (p. 101)

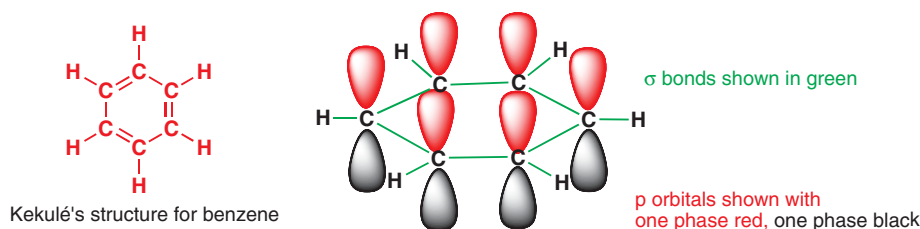


Molecules with more than one C=C double bond

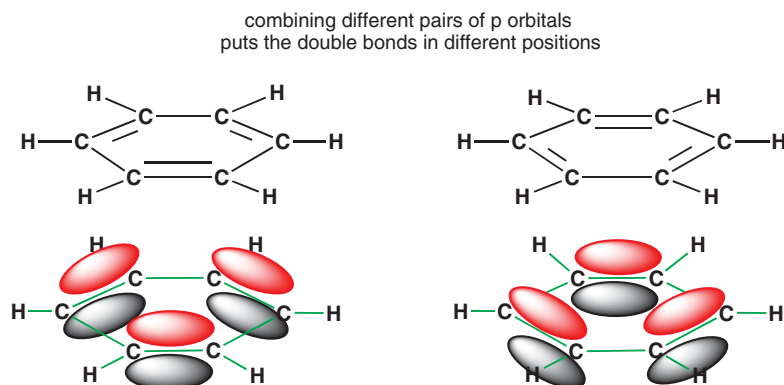
Benzene has three strongly interacting double bonds

The rest of this chapter concerns molecules with more than one C=C double bond and what happens to the π orbitals when they interact. To start, we shall take a bit of a jump and look at the structure of benzene. Benzene has been the subject of considerable controversy since its discovery in 1825. It was soon worked out that the formula was C_6H_6 , but how were these atoms arranged? Some strange structures were suggested until Kekulé proposed the correct structure in 1865.

Shown below are the molecular orbitals for Kekulé's structure. As in simple alkenes, each of the carbon atoms is sp^2 hybridized, leaving the remaining p orbital free.



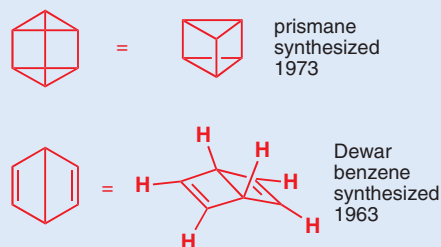
The σ framework of the benzene ring is like the framework of an alkene, and for simplicity we have just represented the σ bonds as green lines. The difficulty comes with the p orbitals—which pairs do we combine to form the π bonds? There seem to be two possibilities.



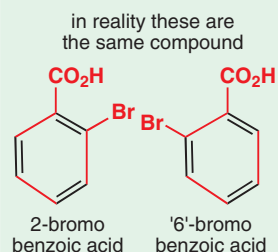
With benzene itself, these two forms are identical but, if we had a 1,2- or a 1,3-disubstituted benzene compound, these two forms would be different. A synthesis was designed for the two compounds in the box on the right but it was found that both compounds were identical. This posed a problem to Kekulé—his structure didn't seem to work after all. His solution—which we now know to be incorrect—was that benzene rapidly equilibrates, or 'resonates', between the two forms to give an averaged structure in between the two.

The molecular orbital answer to this problem is that all six p orbitals can combine to form (six) new molecular orbitals, and the electrons in these orbitals form a ring of electron density above and below the plane of the molecule. Benzene *does not resonate* between the two Kekulé structures—the electrons are in molecular orbitals spread equally over all the carbon atoms. However, the term 'resonance' is still sometimes used (but not in this book) to describe the averaging effect of this mixing of molecular orbitals. We shall describe the π electrons in benzene as **delocalized**, that is, no longer localized in specific double bonds between two particular carbon atoms but spread out, or delocalized, over all six atoms in the ring.

The two early proposals for the structure of benzene were wrong, but nonetheless are stable isomers of benzene (they are both C_6H_6) which have since been synthesized. For more on the Kekulé structure, see p. 24.



■ For example, if the double bonds were localized then these two compounds would be chemically different. (The double bonds are drawn shorter than the single bonds to emphasize the difference.)

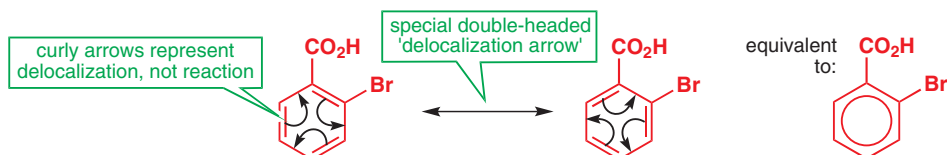




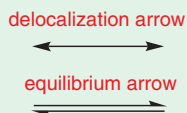
the circle represents the delocalized system

The alternative drawing on the left shows the π system as a ring and does not put in the double bonds: you may feel that this is a more accurate representation, but it does present a problem when it comes to writing mechanisms. As you saw in Chapter 5, the curly arrows we use represent two electrons. The circle here represents *six* electrons, so in order to write reasonable mechanisms we still need to draw benzene *as though* the double bonds were localized. However, when you do so, you must keep in mind that the electrons are delocalized, and it does not matter which of the two arrangements of double bonds you draw.

If we want to represent delocalization using these ‘localized’ structures, we can do so using curly arrows. Here, for example, are the two ‘localized’ structures corresponding to 2-bromocarboxylic acid. The double bonds are not localized, and the relationship between the two structures can be represented with curly arrows which indicate how one set of bonds map onto the other.

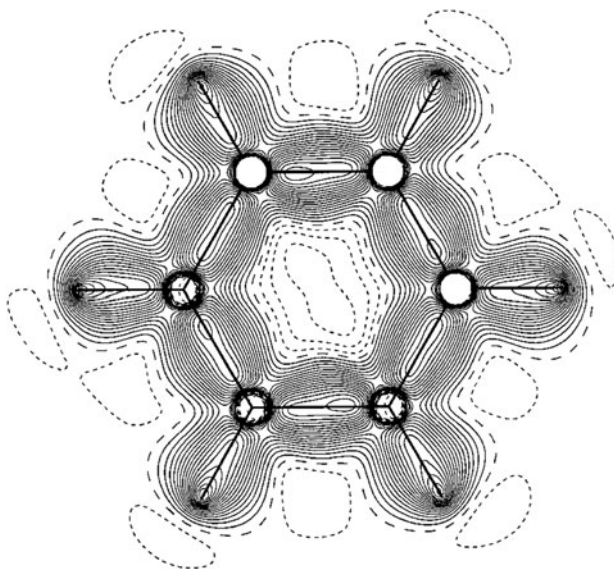


■ The delocalization arrow is used to connect two representations of the same structure. Don't get it confused with an equilibrium arrow, which is used to show two structures interconverting. In an equilibrium, at least one σ bond must change place.



These curly arrows are similar to the ones we introduced in Chapter 5, but there is a crucial difference: here, there is no reaction taking place. In a real reaction, electrons move. Here, they do not: the only things that ‘move’ are the double bonds in the structures. The curly arrows just show the link between alternative representations of exactly the same molecule. You must not think of them as showing ‘movement round the ring’. To emphasize this difference we also use a different type of arrow connecting them—a delocalization arrow made up of a single line with an arrow at each end. Delocalization arrows remind us that our simple fixed-bond structures do not tell the whole truth and that the real structure is a mixture of both.

The fact that *the π electrons are not localized* in alternating double bonds but are *spread out over the whole system* in a ring is supported by theoretical calculations and confirmed by experimental observations. Electron diffraction studies show benzene to be a regular planar hexagon with all the carbon–carbon bond lengths identical (139.5 pm). This bond length is in between that of a carbon–carbon single bond (154.1 pm) and a full carbon–carbon double bond (133.7 pm). A further strong piece of evidence for this ring of electrons is revealed by proton NMR and discussed in Chapter 13.



Electron diffraction image of a molecule of benzene

How to describe delocalization?

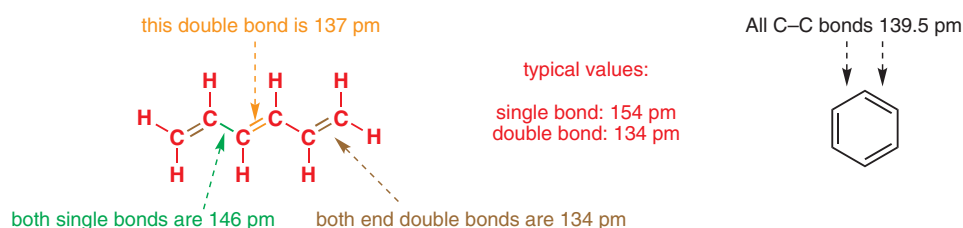
What words should be used to describe delocalization is a vexed question. Terms such as *resonance*, *mesomerism*, *conjugation*, and *delocalization* are only a few of the ones you will find in books. You will already have noticed that we're avoiding *resonance* because it carries a suggestion that the structure is somehow oscillating between localized structures. We shall use the words *conjugation* and *delocalization*: conjugation focuses on the way that double bonds link together into a single π system, while delocalization focuses on the electrons themselves. Adjacent double bonds, as you will see, are *conjugated*; the electrons in them are *delocalized*.

Multiple double bonds not in a ring

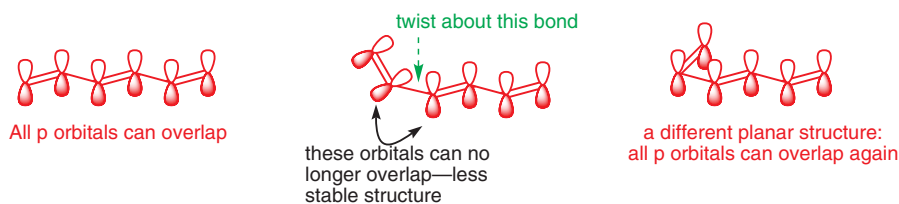
Are electrons still delocalized even when there is no ring? To consider this, we'll look at hexatriene—three double bonds and six carbons, like benzene, but without the ring. There are two isomers of hexatriene, with different chemical and physical properties, because the central double bond can adopt a *cis* or a *trans* geometry. The structures of both *cis*- and *trans*-hexatriene have been determined by electron diffraction and two important features emerge:

- Both structures are essentially planar.
- Unlike benzene, the double and single bonds have different lengths, but the central double bond in each case is slightly longer than the end double bonds and the single bonds are slightly shorter than a 'standard' single bond.

Here's the most stable structure of *trans*-hexatriene, with benzene shown for comparison.



The reason for the deviation of the bond lengths from typical values and the preference for planar structures is again due to the molecular orbitals which arise from the combination of the six p orbitals. Just as in benzene, these orbitals can combine to give one molecular orbital stretching over the whole molecule. The p orbitals can overlap and combine only if the molecule is planar.

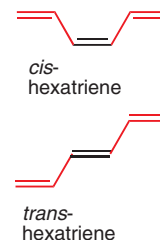


If the molecule is twisted about one of the single bonds, then some overlap is lost, making it harder to twist about the single bonds in this structure than in a simple alkene. Other planar arrangements are stable, however, and *trans*-hexatriene can adopt any of the planar conformations shown in the margin.

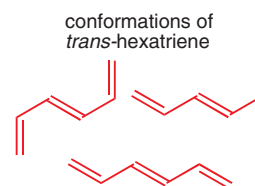
Conjugation

In benzene and hexatriene every carbon atom is sp^2 hybridized with one p orbital available to overlap with its neighbours. The uninterrupted chain of p orbitals is a consequence of having alternate double and single bonds. When two double bonds are separated by just one single bond, the two double bonds are said to be *conjugated*. Conjugated double bonds have different properties from isolated double bonds, both physically (they are often longer, as you have just seen) and chemically (see Chapters 22).

You have already met several conjugated systems: lycopene at the start of this chapter and β -carotene in Chapter 3, for example. Each of the 11 double bonds in β -carotene is separated



■ The terminal double bonds can't have two forms because they have only one substituent.

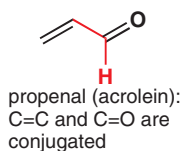


➔ Conformation is the topic of Chapter 16.

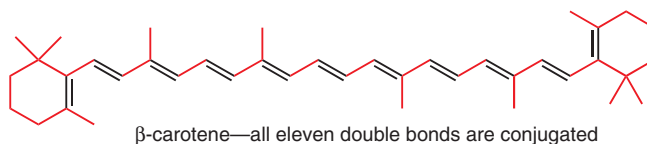
Conjugation

In the dictionary 'conjugated' is defined, among other ways, as 'joined together, especially in pairs' and 'acting or operating as if joined'. This does indeed fit very well with the behaviour of such conjugated double bonds, since the properties of a conjugated system are different from those of the component parts.

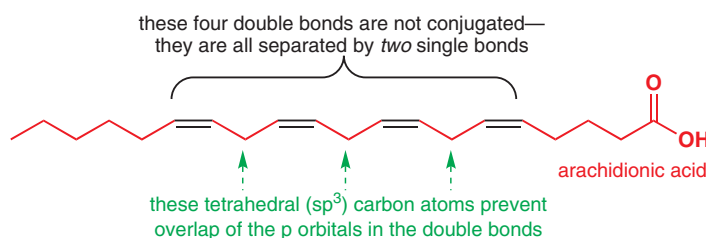
from its neighbour by only one single bond. We again have a long chain in which all the p orbitals can overlap to form molecular orbitals.



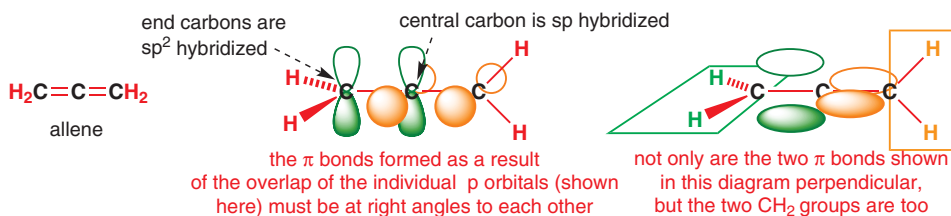
■ The chemistry of such conjugated carbonyl compounds is significantly different from the chemistry of the component parts. The alkene in propenal, for example, is electrophilic and not nucleophilic. This will be explained in Chapter 22.



It is not necessary to have two C=O double bonds in order to have a conjugated system—the C=C and C=O double bonds of propenal (acrolein) are also conjugated. What is important is that the double bonds are separated by *one and only one* single bond. Here's a counter-example: arachidonic acid is one of the fabled 'polyunsaturated' fatty acids. None of the four double bonds in this structure are conjugated since in between any two double bonds there is an sp^3 carbon. This means that there is no p orbital available to overlap with the ones from the double bonds. The saturated carbon atoms 'insulate' the double bonds from each other and prevent conjugation.



If an atom has two double bonds directly attached to it, that is, there are no single bonds separating them, again no conjugation is possible. The simplest compound with such an arrangement is allene. The arrangement of the p orbitals in allene means that no delocalization is possible because the two π bonds are perpendicular to each other.



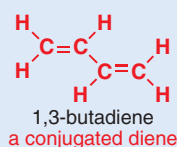
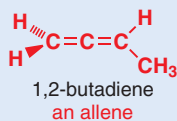
Interactive bonding orbitals in allene

● Requirements for conjugation

- Conjugation requires double bonds separated by one single bond.
- Double bonds separated by two single bonds or no single bonds are not conjugated.

Isomers of butadiene

Butadiene normally refers to 1,3-butadiene. It is also possible to have 1,2-butadiene, which is another example of an allene.



The conjugation of two π bonds

To understand the effects of conjugation on molecules, we need now to look at their molecular orbitals. We'll concentrate only on the electrons in π orbitals—you can take it that all the C–C and C–H σ bonds are essentially the same as those of all the other molecules you met in Chapter 4. We'll start with the simplest compound that can have two conjugated π bonds: butadiene. As you would expect, butadiene prefers to be planar to maximize overlap between its p orbitals. But exactly how does that overlap happen, and how does it give rise to bonding?

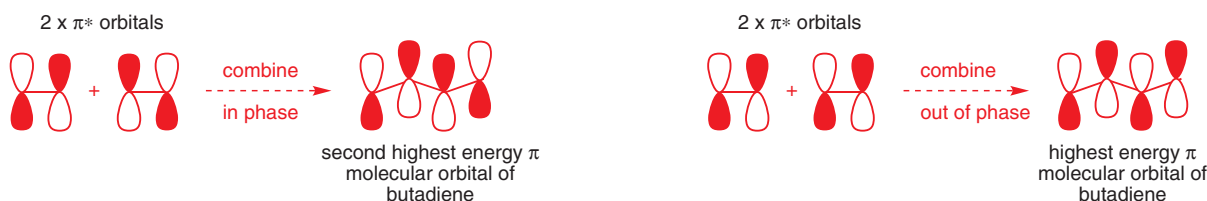
The molecular orbitals of butadiene

Butadiene has two π bonds, each made up of two p orbitals: a total of four atomic orbitals. We'd therefore expect four molecular orbitals, housing four electrons. Just like

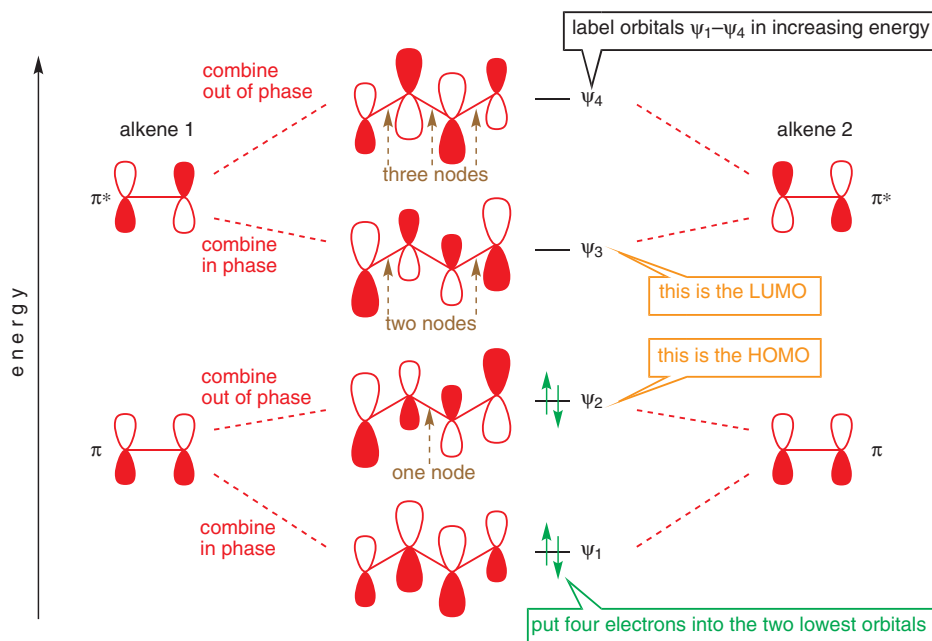
hexatriene above, these orbitals extend over the whole molecule, but we can easily work out what these molecular orbitals look like simply by taking the orbitals of two alkenes and interacting them side by side. We have two π orbitals and two π^* orbitals, and we can interact them in phase or out of phase. Here are the first two, made by interacting the two π orbitals:



and the next two, made from two π^* orbitals:



We can represent all four molecular orbitals like this, stacked up in order of their energy in a molecular orbital energy level diagram. With four orbitals, we can't just use '*' to represent antibonding orbitals, so conventionally they are numbered ψ_1 – ψ_4 (ψ is the Greek letter psi).



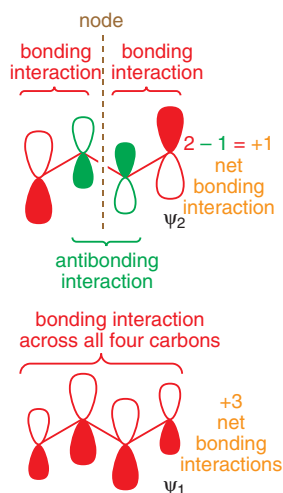
It's worth noticing a couple of other things about the way we have represented these four molecular orbitals before we move on. Firstly, the number of nodes (changes in phase as you move from one orbital to the next) increases from zero in ψ_1 to three in ψ_4 . Secondly, notice that the p orbitals making up the π system are not all shown as the same size—their *coefficients* vary according to the orbital they are in. This is a mathematical consequence of the way the orbitals sum together, and you need not be concerned with the details, just the general principle that ψ_1 and ψ_4 have the largest coefficients in the middle; ψ_2 and ψ_3 the largest coefficients at the ends.

Now for the electrons: each orbital holds two electrons, so the four electrons in the π system go into orbitals ψ_1 and ψ_2 .

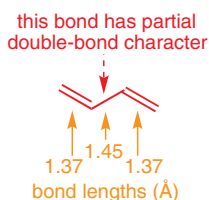
Interactive bonding orbitals in butadiene

➔ The idea that higher energy orbitals have more nodes is familiar to you from Chapter 4—see p. 88.

The term 'coefficient' describes the contribution of an individual atomic orbital to a molecular orbital. It is represented by the size of the lobes on each atom.



■ In our glimpse of hexatriene earlier in this chapter we saw similar effects: a tendency to be planar and restriction to rotation about the slightly shortened single bonds.



it takes 30 kJ mol⁻¹ to rotate about this bond

it takes 3 kJ mol⁻¹ to rotate about this single bond

rotation about a full double bond needs more than 260 kJ mol⁻¹: it's essentially impossible

The molecular orbital diagram also helps us explain some aspects of the reactivity of butadiene. Notice that we have marked on for you the HOMO (ψ₂) and the LUMO (ψ₃). On either side you can see the equivalent HOMO (π orbital) and LUMO (π* orbital) for the isolated alkene (i.e. ethene). Some relevant features to note:

- The overall energy of the two bonding butadiene molecular orbitals is lower than that of the two molecular orbitals for ethene. This means that conjugated butadiene is more thermodynamically stable than just two isolated double bonds.
- The HOMO of butadiene is *higher* in energy than the HOMO for ethene. This is consistent with the fact that butadiene is *more* reactive than ethene towards electrophiles.
- The LUMO for butadiene is *lower* in energy than the LUMO for ethene. This is consistent with the fact that butadiene is *more* reactive than ethene towards nucleophiles.

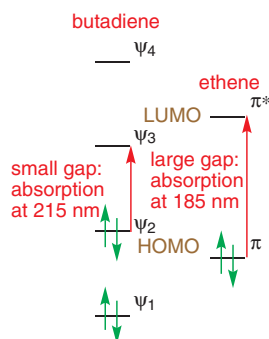
So conjugation makes butadiene more stable, but it also makes it more reactive to both nucleophiles and electrophiles! This superficially surprising result is revisited in detail in Chapter 19.

UV and visible spectra

In Chapter 2 you saw how, if given the right amount of energy, electrons can be promoted from a low-energy atomic orbital to a higher energy one and how this gives rise to an atomic absorption spectrum. Exactly the same process can occur with molecular orbitals: energy of the right wavelength can promote an electron from a filled orbital (for example the HOMO) to an unfilled one (for example the LUMO), and plotting the absorption of energy against wavelength gives rise to a new type of spectrum called, for obvious reasons which you will see in a moment, a UV-visible spectrum.

You have just seen that the energy difference between the HOMO and LUMO for butadiene is less than that for ethene. We would therefore expect butadiene to absorb light of longer

■ To understand this section well you will need to remember the formulae linking energy and wavelength, $E = h\nu$, and energy and frequency, $E = hc/\lambda$. See p. 53 for more on these.

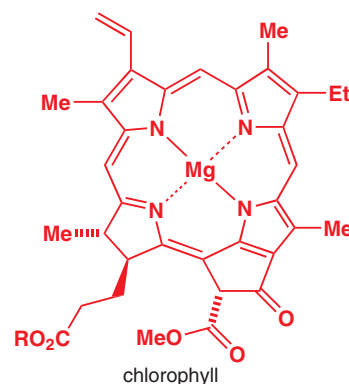
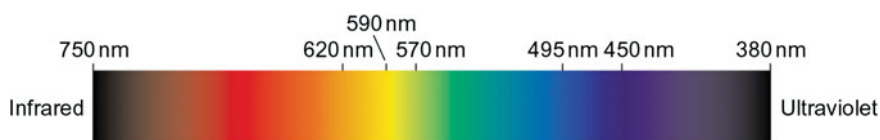


wavelength than ethene (the longer the wavelength the lower the energy). This is indeed the case: butadiene absorbs at 215 nm compared to 185 nm for ethene. The conjugation in butadiene means it absorbs light of a longer wavelength than ethene. One of the consequences of conjugation is to lessen the gaps between filled and empty orbitals, and so allow absorption of light of a longer wavelength.

UV absorption in ethene and butadiene

● The more conjugated a compound is, the smaller the energy transition between its HOMO and LUMO, and hence the longer the wavelength of light it can absorb. UV-visible spectroscopy can tell us about the conjugation present in a molecule.

Both ethene and butadiene absorb in the UV region of the electromagnetic spectrum. If we extend the conjugation further, the gap between HOMO and LUMO will eventually be small enough to allow the compound to absorb visible light and hence have a colour. Lycopene, the pigment in tomatoes, which we introduced at the start of the chapter, has 11 conjugated double bonds (plus two unconjugated ones). It absorbs blue-green light at about 470 nm: consequently tomatoes are red. Chlorophyll, in the margin, has a cyclic conjugated system: it absorbs at long wavelengths and is green.



The colour of pigments depends on conjugation

It is no coincidence that these and many other highly conjugated compounds are coloured. All dyes and pigments based on organic compounds are highly conjugated.

The table below shows the approximate wavelengths of light absorbed by a polyene conjugated system containing various numbers n of double bonds. Note that the colour absorbed is complementary to the colour transmitted—a red compound must absorb blue and green light to appear red.

Approximate wavelengths for different colours

Absorbed frequency, nm	Colour absorbed	Colour transmitted	$R(\text{CH}=\text{CH})_n\text{R}$, $n =$
200–400	ultraviolet	—	< 8
400	violet	yellow–green	8
425	indigo–blue	yellow	9
450	blue	orange	10
490	blue–green	red	11
510	green	purple	
530	yellow–green	violet	
550	yellow	indigo–blue	
590	orange	blue	
640	red	blue–green	
730	purple	green	

Fewer than about eight conjugated double bonds, and the compound absorbs only in the UV. With more than eight conjugated double bonds, the absorption creeps into the visible and, by the time it reaches 11, the compound is red. Blue or green polyenes are rare, and dyes of these colours rely on more elaborate conjugated systems.

In colour chemistry, *dye* is a soluble colorant while a *pigment* is made of insoluble coloured particles. In biology the word pigment is used for any coloured compound. Dyeing pigments are often inorganic compounds, which are coloured for reasons other than conjugation, but nonetheless to do with the gaps between orbitals.

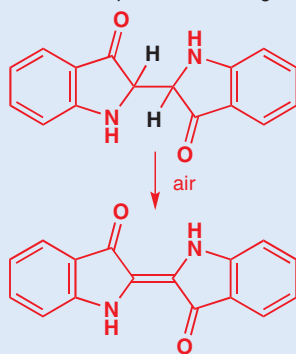
UV absorption in linear conjugated polyenes

Blue jeans

Transitions from bonding to antibonding π orbitals are called $\pi \rightarrow \pi^*$ transitions. If electrons are instead promoted from a non-bonding lone pair (n orbital) to a π^* orbital (an $n \rightarrow \pi^*$ transition) smaller energy gaps may be available, and many dyes make use of $n \rightarrow \pi^*$ transitions to produce colours throughout the whole spectrum. For example, the colour of blue jeans comes from the pigment indigo. The two nitrogen atoms provide the lone pairs that can be excited into the π^* orbitals of the rest of the molecule. These are low in energy because of the two carbonyl groups. Yellow light is absorbed by this pigment and indigo–blue light is transmitted.

Jeans are dyed by immersion in a vat of reduced indigo, which is colourless since the conjugation is interrupted by the central single bond. When the cloth is hung up to dry, the oxygen in the air oxidizes the pigment to indigo and the jeans turn blue.

colourless precursor to indigo



indigo: the pigment of blue jeans



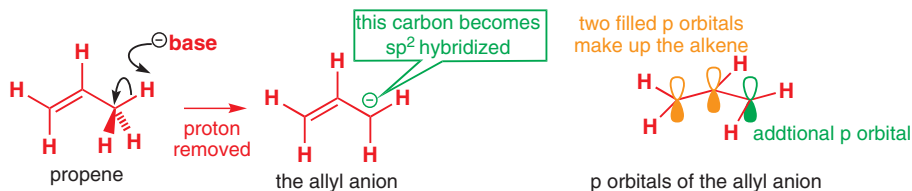
The allyl system

The allyl anion

In butadiene, four atomic p orbitals interact to make four molecular orbitals; in hexatriene (and you will soon see benzene too) six atomic orbitals interact to make six molecular orbitals. We are now going to consider some common conjugated systems made up of *three* interacting p orbitals. We'll start with the structure we get from treating propene with a very strong base—one strong enough to remove one of the protons from its methyl group. H^+ is removed, so the product must have a negative charge, which formally resides on the carbon of what was the methyl group. That carbon atom started off sp^3 hybridized (i.e. tetrahedral: it had four substituents), but after it has been deprotonated it must become trigonal (sp^2), with only three substituents plus a p orbital to house the negative charge.

➡ You will meet such super-strength bases in the next chapter.

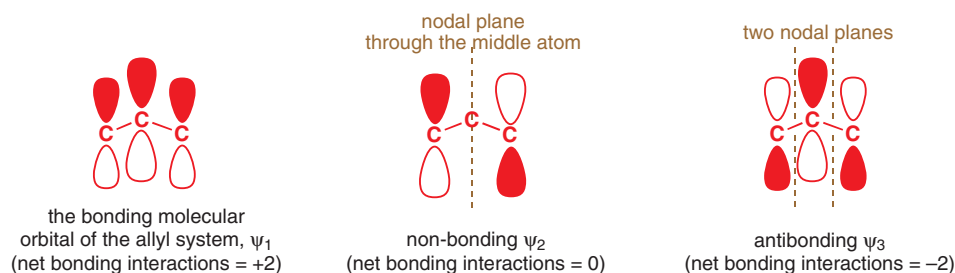
■ Of course the anion doesn't really exist 'free' like this; it will most likely have a metal cation to which it is coordinated in some way. The arguments we are going to apply about its structure are still valid whether or not there is a metal associated with it.



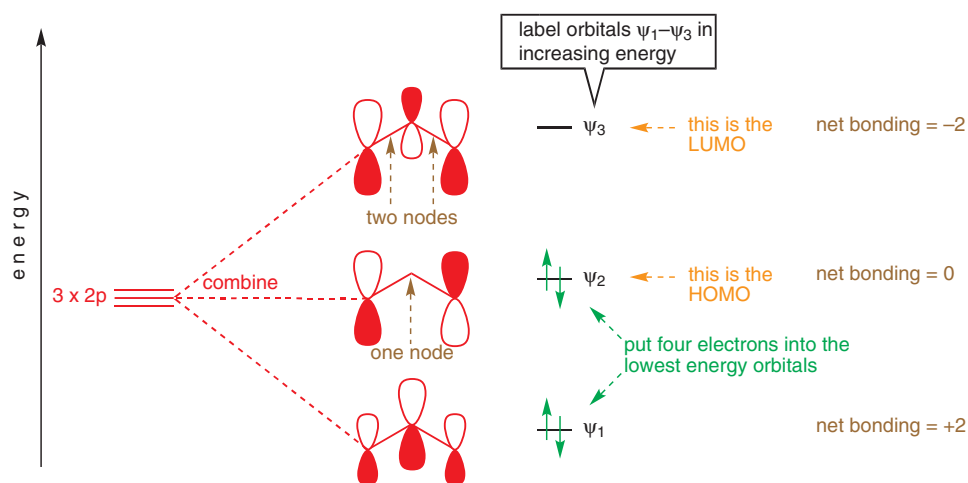
We could work out the orbitals of the allyl anion by combining this p orbital with a ready-made π bond, but instead this time we will start with the three separate p atomic orbitals and combine them to get three molecular orbitals. At first we are not concerned about where the electrons are—we are just building up the molecular orbitals.

The lowest energy orbital (ψ_1) will have them all combining in phase. This is a bonding orbital since all the interactions are bonding. The next orbital (ψ_2) requires one node, and the only way to include a node and maintain the symmetry of the system is to put the node through the central atom. This means that when this orbital is occupied there will be no electron density on this central atom. Since there are no interactions between adjacent atomic orbitals (either bonding or antibonding), this is a non-bonding orbital. The final molecular

orbital (ψ_3) must have two nodal planes. All the interactions of the atomic orbitals are out of phase so the resulting molecular orbital is an antibonding orbital.



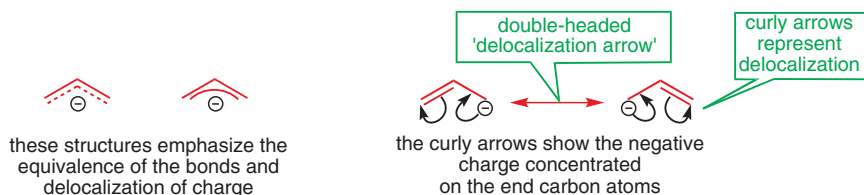
We can summarize all this information in a molecular orbital energy level diagram, and at the same time put the electrons into the orbitals. We need four electrons—two from the alkene π bond and two more for the anion (these were the two in the C–H bond, and they are still there because only a proton, H^+ , was removed). The four electrons go into the lowest two orbitals, ψ_1 and ψ_2 , leaving ψ_3 vacant. Notice too that the energy of two of the electrons is lower than it would have been if they had remained in unconjugated p orbitals: conjugation lowers the energy of filled orbitals and makes compounds more stable.



■ This diagram shows only the π orbitals of the allyl system. We have ignored all the molecular orbitals from the σ framework because the bonding σ orbitals are considerably lower in energy than the molecular orbitals of the π system and the vacant antibonding σ^* orbitals are much higher in energy than the π antibonding molecular orbital.

Interactive bonding orbitals in the allyl anion

Where is the electron density in the allyl anion π system? We have two filled π molecular orbitals and the electron density comes from a sum of both orbitals. This means there is electron density on all three carbon atoms. However, the coefficients of the end carbons are of a significant size in both orbitals, but in ψ_2 the middle carbon has no electron density at all—it lies on a node. So overall, even though the negative charge is spread over the whole molecule, the end carbons carry more of the electron density than the middle one. We can represent this in two ways—the first structure below emphasizes the delocalization of the charge over the whole molecule, but fails to get across the important point that the negative charge resides principally at the ends. Curly arrows do this much better: we can use them to show that the negative charge is not localized, but principally divided between the two end carbons.



■ A reminder: this is not an equilibrium—the arrows do not represent the movement of charge. The two structures are alternative, imperfect representations of an 'averaged' structure, and they are linked by a double-headed delocalization arrow.

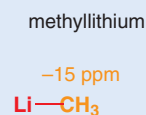
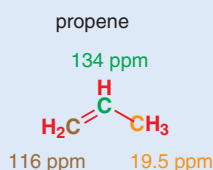
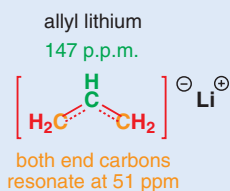


The problem with these structures carrying curly arrows is that they seem to imply that the negative charge (and the double bond for that matter) is jumping from one end of the molecule to the other. This, as we have seen, is just not so. Another and perhaps better picture uses dotted lines and partial charges. But the structure with the dotted bonds, as with the representation of benzene with a circle in the middle, is no good for writing mechanisms. Each of the representations has its strong and weak points: we shall use each as the occasion demands.

Using NMR to study delocalization

Delocalization of the allyl anion, and the localization of the negative charge mainly on the end carbons, is clear from its ^{13}C NMR spectrum as well. In Chapter 3 we explained that ^{13}C NMR gives us a good measure of the amount of electron density around a C atom—the extent to which it is deshielded and therefore exposed to the applied magnetic field. If you need reminding about the terminology, theory, and practice of NMR, turn back now to Chapter 3, pp. 52–63.

It is possible to record a ^{13}C NMR spectrum of an allyl anion with a lithium counterion. The spectrum shows only two signals: the middle carbon at 147 ppm and the two end carbons both at 51 ppm. This confirms two things: (i) both end carbons are the same and the structure is delocalized, and (ii) most of the negative charge is on the end carbons—they are more shielded (have a smaller chemical shift) as a result of the greater electron density. In fact, the central carbon's shift of 147 ppm is not far from that of a normal double-bond carbon (compare the signals in propene). The end carbons' shift is in between that of a double bond and a saturated carbon directly bonded to a metal (e.g. methyllithium, whose negative chemical shift results from the highly polarized Li–C bond).

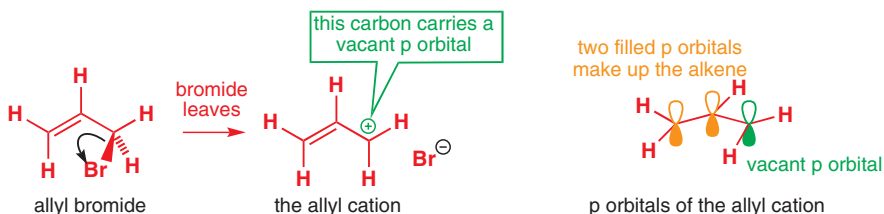


The allyl cation

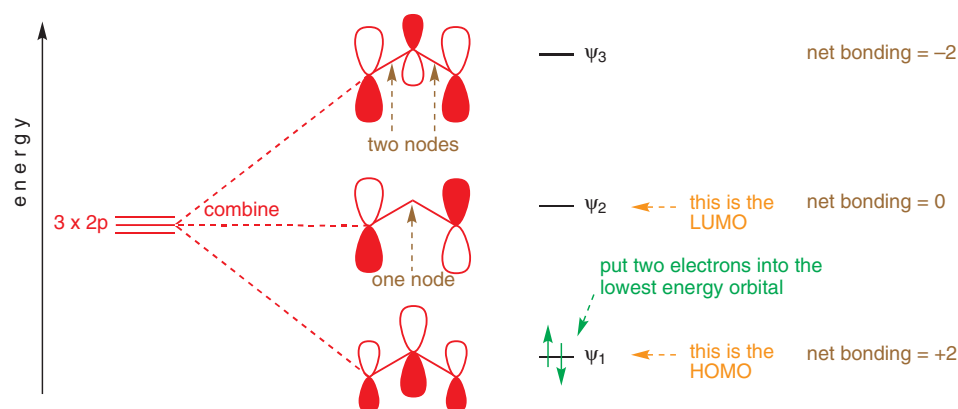


What if, instead of taking just a proton, we had also taken away two electrons from propene? In reality we can get such a structure quite straightforwardly from allyl bromide (prop-2-enyl bromide or 1-bromoprop-2-ene). Carbon 1 in this compound has four atoms attached to it (a carbon, two hydrogens, and a bromine atom) so it is tetrahedral (or sp^3 hybridized).

Bromine is more electronegative than carbon and so the C–Br bond is polarized towards the bromine. It is quite easy to break this bond completely, with the bromine keeping both electrons from the C–Br bond to become bromide ion, Br^- , leaving behind an *allyl cation*. The positively charged carbon now has only three substituents so it becomes trigonal (sp^2 hybridized). It must therefore have a vacant p orbital.



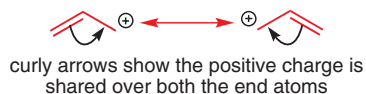
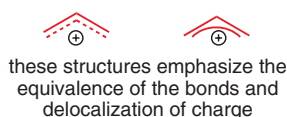
Like the allyl anion, the orbitals in the allyl cation are a combination of three atomic p orbitals, one from each carbon. So we can use the same molecular orbital energy level diagram as we did for the anion, simply by adjusting the number of electrons we put into the orbitals. This time, there are only two electrons, from the alkene, as those which were in the C–Br bond have left with anionic bromide.



The two electrons in the filled orbital are in a lower energy orbital than they would have been if they had stayed in an unconjugated p orbital: as with the anion, *conjugation* leads to *stabilization*.

The two electrons are spread over three carbon atoms. Overall, the allyl cation has a positive charge. But where is the positive charge concentrated? What we need to do is look to see where there is a *deficit* of charge. The only orbital with any electrons in it is the bonding molecular orbital ψ_1 . From the relative sizes of the coefficients on each atom we can see that the middle carbon has more electron density on it than the end ones, so the end carbons must be more positive than the middle one.

We expect both end carbons to be identical, and ^{13}C NMR tells us that this is so (see below). Again we need a way of showing this delocalization, either on a single structure or as a pair of localized structures linked by a delocalization arrow.

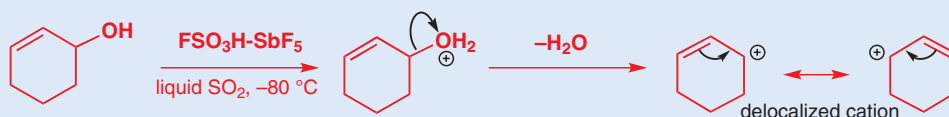


Notice how we draw the curly arrows here: we want to show the positive charge ‘moving’, and it is tempting to draw a curly arrow starting from the positive charge. But curly arrows must always start on something representing a pair of electrons. So we must move the positive charge as a consequence of the movement of the electrons in the double bond: as we pull them away from one end, they leave behind a positive charge.

➔ The guidelines for drawing curly arrows are given on p. 123.

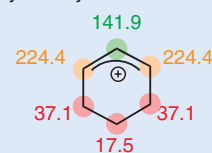
The NMR spectrum of a delocalized cation

The reaction below shows the formation of a cation close in structure to the allyl cation. A very strong acid (called ‘super-acid’—see Chapter 15) protonates the OH group of 3-cyclohexenol, which can then leave as water. The resulting cation is, not surprisingly, unstable and would normally react rapidly with a nucleophile. However, at low temperatures and if there are no nucleophiles present, the cation is relatively stable and it is even possible to record a ^{13}C NMR spectrum (at -80°C).



The ^{13}C NMR spectrum of this allylic cation reveals a plane of symmetry, which confirms that the positive charge is spread over two carbons. The large shift of 224 ppm for these carbons indicates very strong deshielding (that is, lack of electrons) but is nowhere near as large as that of a localized cation (which would resonate at about 330 ppm). The middle carbon’s shift of 142 ppm is almost typical of a normal double bond, indicating that it is neither significantly more nor less electron-rich than normal. In fact, it is interesting to note that the middle carbon of this cation and the allyl anion we described above have almost exactly the same chemical shift—proof that the charge lies mainly on the ends of the allyl system.

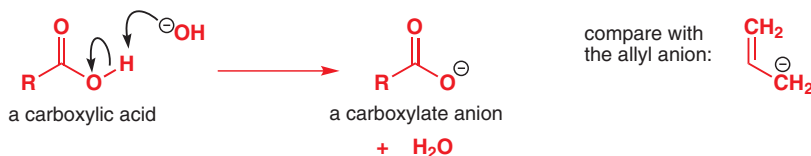
^{13}C NMR shifts in ppm—notice the plane of symmetry down the middle



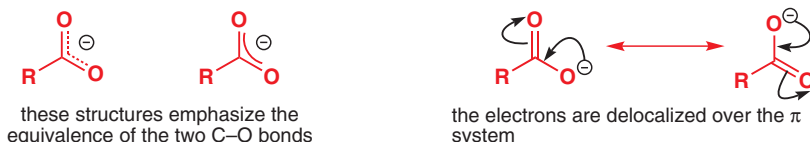
Delocalization over three atoms is a common structural feature

The carboxylate anion

You may already be familiar with one anion very much like the allyl anion—the carboxylate ion, which forms when a carboxylic acid reacts with a base. In this structure we again have a negatively charged atom separated from a double bond by a single bond adjacent to a single bond: it's analogous to an allyl anion with oxygen atoms replacing two of the carbon atoms.

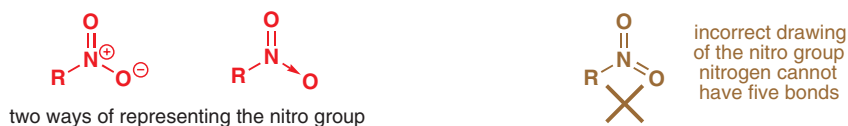


X-ray crystallography shows both carbon–oxygen bond lengths in this anion to be the same (136 pm), in between that of a normal carbon–oxygen double bond (123 pm) and single bond (143 pm). The negative charge is spread out equally over the two oxygen atoms, and we can represent this in two ways—as before, the one on the left shows the equivalence of the two C–O bonds, but you would use the one on the right for writing mechanisms. The delocalization arrow tells us that both localized forms contribute to the real structure.



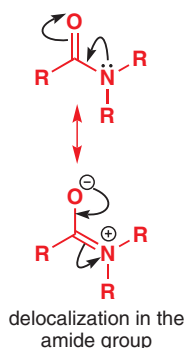
The nitro group

The nitro group consists of a nitrogen bonded to two oxygen atoms and a carbon (for example an alkyl group). There are two ways of representing the structure: one using formal charges, the other (which we suggest you avoid) using a dative bond. Notice in each case that one oxygen is depicted as being doubly bonded, the other singly bonded. Drawing both oxygen atoms doubly bonded is incorrect—*nitrogen cannot participate in five bonds*: this would require ten bonding electrons around the N atom, and there are not enough s and p orbitals to put them in.



➡ This important principle was explained on p. 30.

We call structures such as a nitro and a carboxylate group *isoelectronic*: the atoms may be different but the number of and arrangement of the bonding electrons are the same.



The problem even with the 'correct' structure on the left is that the equivalence of the two N–O bonds is not made clear. The nitro group has exactly the same number of electrons as a carboxylate anion (although it's neutral of course because nitrogen already has one more electron than carbon) and the delocalized structure can be shown with curly arrows in the same way.

We have not shown molecular orbital energy level diagrams for the carboxylate and nitro groups, since they are similar to that of the allyl anion. Only the absolute energies of the molecular orbitals are different since different elements with different electronegativities are involved in each case.

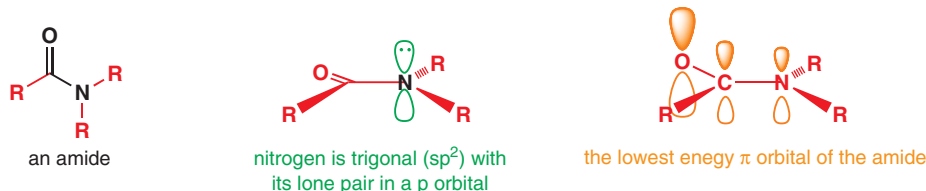
The amide group

Life is built of amides, because the amide group is the link through which amino acids join together to form the proteins which make up much of the structural features of living systems.

Nylon is a synthetic polyamide, and shares with many proteins the property of durability. The structure of this deceptively simple functional group has an unexpected feature which is responsible for much of the stability it confers.

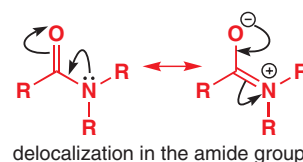
The allyl anion, carboxylate, and nitro groups have four electrons in a π system spread out over three atoms. The nitrogen in the amide group also has a pair of electrons that can conjugate with the π bond of the carbonyl group. For effective overlap with the π bond, this lone pair of electrons must be in a p orbital. This in turn means that the nitrogen must be sp^2 hybridized.

■ Contrast this with the lone pair of a typical amine, which lies in an sp^3 orbital (see p. 103): an amine N is pyramidal (sp^3) while an amide N is trigonal planar (sp^2).

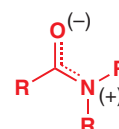


In the carboxylate ion, a negative charge is shared (equally) between two oxygen atoms. In an amide there is no charge as such—the lone pair on nitrogen is shared between the nitrogen and the oxygen. The delocalization can be shown as usual by using curly arrows, as shown in the margin.

This representation suffers from the usual problems. Curly arrows usually show electron movement, but here they do not: they simply show how to get from one of the alternative representations to the other. The molecular orbital picture of the amide tells us that the electrons are unevenly distributed over the three atoms in the π system with a greater electron density on the oxygen: you can see this in the delocalized structure on the right, which has a full negative charge on O and a positive charge on N. (We also indicated this in the diagram of the lowest energy π orbital above, which has a greatest coefficient, and therefore greatest electron density, on O.) Another aspect of the structure of the amide group that this pair of structures indicates correctly is that there is partial double bond character between the C atom and the N atom. We will come back to this shortly.



The real structure of the amide group lies in between the two extreme structures linked by the delocalization arrow: a better representation might be the structure on the right. The charges in brackets indicate substantial, although not complete, charges, maybe about a half plus or minus charge. However, we cannot draw mechanisms using this structure.

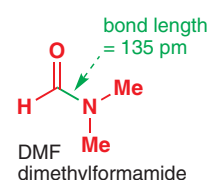


We can summarize several points about the structure of the amide group, and we will then return to each in a little more detail

- The amide group is planar—this includes the first carbon atoms of the R groups attached to the carbonyl group and to the nitrogen atom.
- The lone pair of electrons on nitrogen is delocalized into the carbonyl group.
- The C–N bond is strengthened by this interaction—it takes on partial double bond character. This also means that we no longer have free rotation about the C–N bond, which we would expect if it were only a single bond.
- The oxygen is more electron-rich than the nitrogen. Hence we might expect the oxygen rather than the nitrogen to be the site of electrophilic attack.
- The amide group as a whole is made more stable as a result of the delocalization.

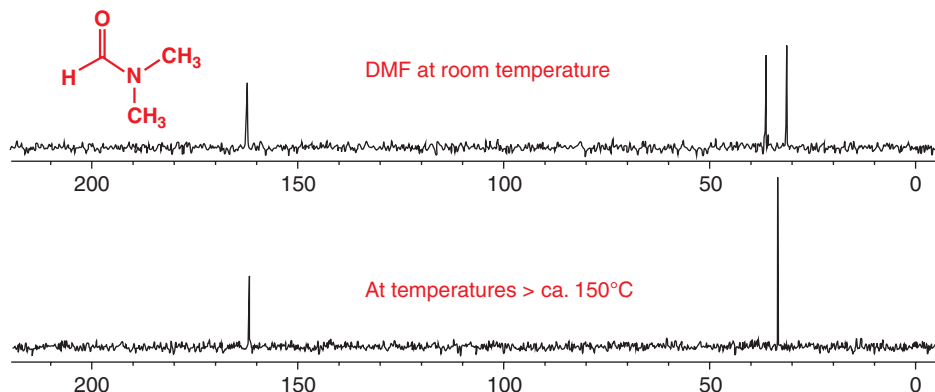
How do we know the amide group is planar? X-ray crystal structures are the simplest answer. Other techniques such as electron diffraction also show that simple (non-crystalline) amides have planar structures. *N,N*-Dimethylformamide (DMF) is an example.

The N–CO bond length in DMF (135 pm) is closer to that of a standard C–N double bond (127 pm) than to that of a single bond (149 pm). This partial double bond character, which the delocalized structures led us to expect, is responsible for restricted rotation about this C–N bond. We must supply 88 kJ mol⁻¹ if we want to rotate the C–N bond in DMF (remember a single bond only takes about 3 kJ mol⁻¹, while a full C–C double bond takes about 260 kJ mol⁻¹). The amount of energy available at room temperature is only enough to allow this bond



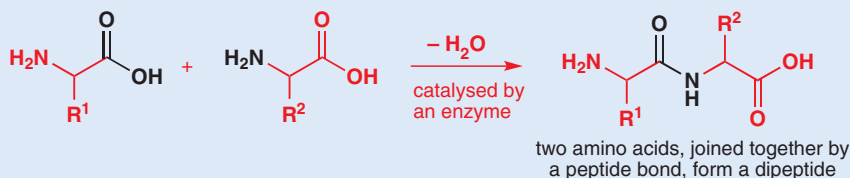
to rotate slowly, and the result is quite clear in the ^{13}C NMR spectrum of DMF. There are three carbon atoms altogether and three signals appear—the two methyl groups on the nitrogen are different. If free rotation were possible about the C–N bond, we would expect to see only two signals, since the two methyl groups would become identical.

■ In fact, if we record the spectrum at higher temperatures, we do indeed only see two signals since now there is sufficient energy available to overcome the rotational barrier and allow the two methyl groups to interchange.



Amides in proteins

Proteins are composed of many amino acids joined together with amide bonds. The amino group of one can combine with the carboxylic acid group of another to give an amide known as a peptide—two amino acids join to form a dipeptide; many join to give a polypeptide.



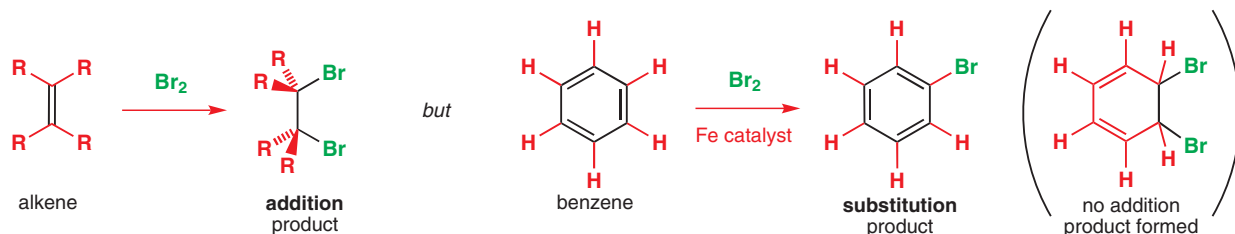
The peptide unit so formed is a planar, rigid structure because of restricted rotation about the C–N bond. This rigidity confers organizational stability on protein structures.

Conjugation and reactivity: looking forward to Chapter 10

Just as delocalization stabilizes the allyl cation and anion (at least some of the electrons in conjugated systems end up in lower energy orbitals than they would have done without conjugation) so too is the amide group stabilized by the conjugation of the nitrogen's lone pair with the carbonyl group. This makes an amide C=O one of the least reactive carbonyl groups (we shall discuss this in Chapter 10). Furthermore, the nitrogen atom of an amide group is very different from that of a typical amine. Most amines are easily protonated. However, since the lone pair on the amide's nitrogen is conjugated into the π system, it is less available for protonation or, indeed, reaction with any electrophile. As a result, when an amide is protonated (and it is not protonated easily, as you will see in the next chapter) it is protonated on oxygen rather than nitrogen. The consequences of conjugation for reactivity extend far and wide, and will be a running theme through many chapters in this book.

Aromaticity

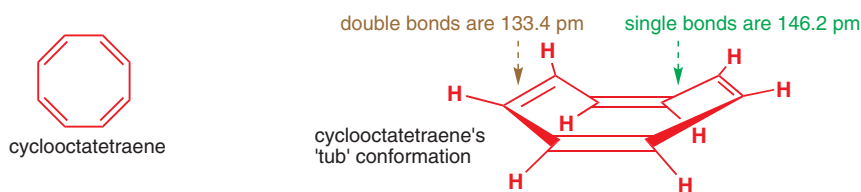
It's now time to go back to the structure of benzene. Benzene is unusually stable for an alkene and is not normally described as an alkene at all. For example, whereas normal alkenes (whether conjugated or not) readily react with bromine to give dibromoalkane *addition* products, benzene reacts with bromine only with difficulty—it needs a catalyst (iron will do) and then the product is a *monosubstituted* benzene and not an addition compound.



Bromine reacts with benzene in a substitution reaction (a bromine atom replaces a hydrogen atom), *keeping the benzene structure intact*. This ability to retain its conjugated structure through all sorts of chemical reactions is one of the important differences between benzene and other alkenes.

What makes benzene special?

You might assume benzene's special feature is its ring structure. To see whether this is the case, we'll look at another cyclic polyene, cyclooctatetraene, with four double bonds in a ring. Given what we have explained about the way that π systems gain stability by allowing overlap between their p orbitals, you may be surprised to find that cyclooctatetraene, unlike benzene, is *not* planar. There is no conjugation between any of the double bonds—there are indeed alternate double and single bonds in the structure, but conjugation is possible only if the p orbitals of the double bonds can overlap and here they do not. The fact that there is no conjugation is shown by the alternating C–C bond lengths in cyclooctatetraene—146.2 and 133.4 pm—which are typical for single and double C–C bonds. If possible, make a model of cyclooctatetraene for yourself—you will find the compound naturally adopts the shape on the right below. This shape is often called a 'tub'.



Interactive structures of cyclooctatetraene, the dianion and dication

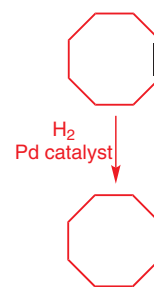
Chemically, cyclooctatetraene behaves like an alkene, not like benzene. With bromine, for example, it forms an addition product and not a substitution product. So benzene is not special just because it is cyclic—cyclooctatetraene is cyclic too but does not behave like benzene.

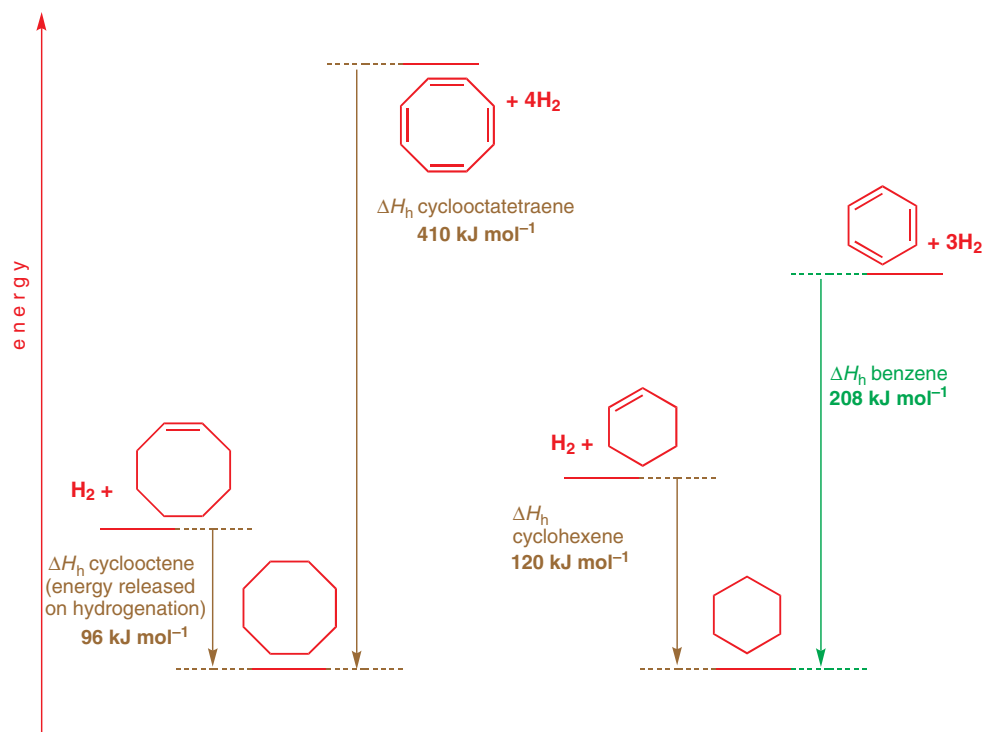
Heats of hydrogenation of benzene and cyclooctatetraene

C=C double bonds can be reduced using hydrogen gas and a metal catalyst (usually nickel or palladium) to produce fully saturated alkanes. This process is called hydrogenation and it is exothermic (that is, energy is released) since a thermodynamically more stable product, an alkane, is produced.

When *cis*-cyclooctene is hydrogenated to cyclooctane, 96 kJ mol⁻¹ of energy is released. Cyclooctatetraene releases 410 kJ mol⁻¹ on hydrogenation. This value is approximately four times one double bond's worth, as we might expect. However, whereas the heat of hydrogenation for cyclohexene is 120 kJ mol⁻¹, on hydrogenating benzene only 208 kJ mol⁻¹ is given out, which is much less than the 360 kJ mol⁻¹ that we would have predicted by multiplying the figure for cyclohexene by 3. Benzene has something to make it stable which cyclooctatetraene does not have.

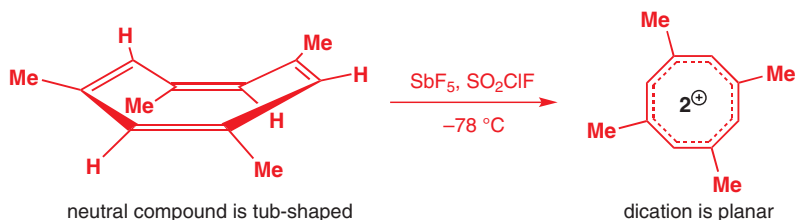
More on this in Chapter 23.





Varying the number of electrons

The mystery deepens when we look at what happens when we treat cyclooctatetraene with powerful oxidizing or reducing agents. If 1,3,5,7-tetramethylcyclooctatetraene is treated at low temperature (-78°C) with $\text{SbF}_5/\text{SO}_2\text{ClF}$ (strongly oxidizing conditions) a dication is formed. This cation, unlike the neutral compound, is *planar* and all the C–C bond lengths are the same.



Interactive structures of cyclooctatetraene, the dianion and dication

This dication still has the same number of atoms as the neutral species, only fewer electrons. The electrons have come from the π system, which is now two electrons short. We could draw a structure showing two localized positive charges, but the charge is in fact spread over the whole ring.

It is also possible to *add* electrons to cyclooctatetraene by treating it with alkali metals and a *dianion* results. X-ray structures reveal this dianion to be planar, again with all C–C bond lengths the same (140.7 pm). The difference between the anion and cation of cyclooctatetraene on the one hand and cyclooctatetraene on the other is the number of electrons in the π system. Neutral, non-planar, cyclooctatetraene has eight π electrons, the planar dication has six π electrons (as does benzene), and the planar anion has ten.

Can you see a pattern forming? The important point is not the number of conjugated atoms but the *number of electrons in the π system*.

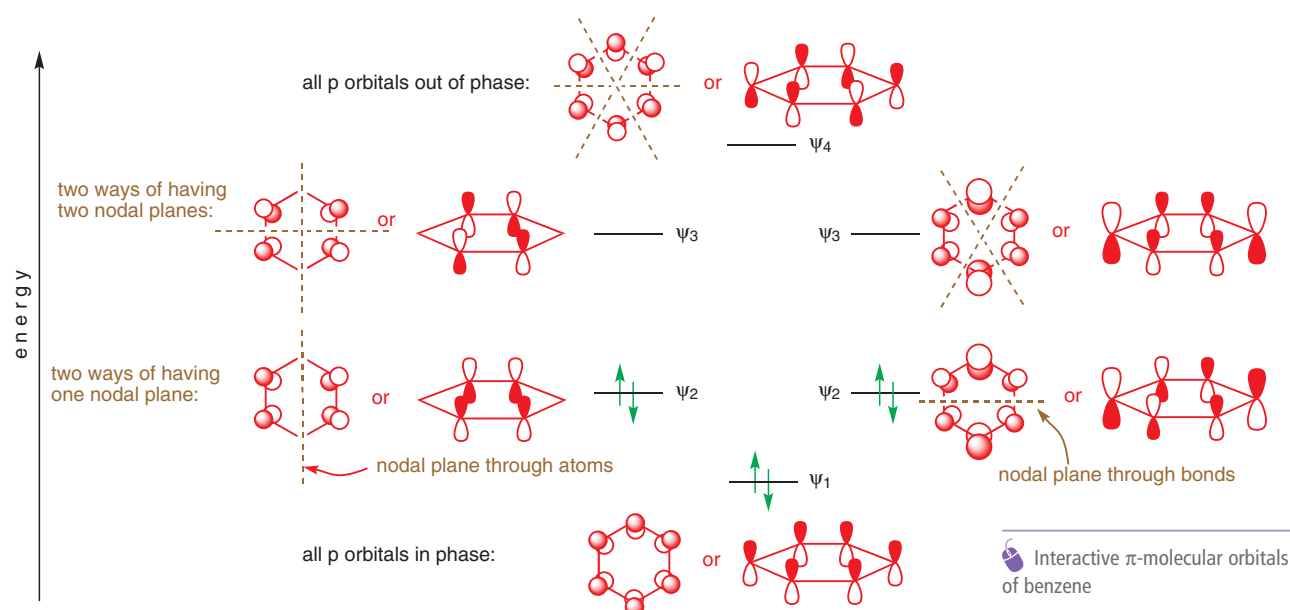
- When they have four or eight π electrons, both six- and eight-membered rings adopt non-planar structures; when they have six or ten π electrons, a planar structure is preferred.

If you made a model of cyclooctatetraene, you might have tried to force it to be flat. If you managed this you probably found that it didn't stay like this for long and that it popped back into the tub shape. The strain in planar cyclooctatetraene can be overcome by the molecule

adopting the tub conformation. The strain is due to the numbers of atoms and double bonds in the ring—it has nothing to do with the number of electrons. The planar dication and dianion of cyclooctatetraene still have this strain. The fact that these ions do adopt planar structures must mean there is some other form of stabilization that outweighs the strain of being planar. This extra stabilization is called *aromaticity*.

Benzene has six π molecular orbitals

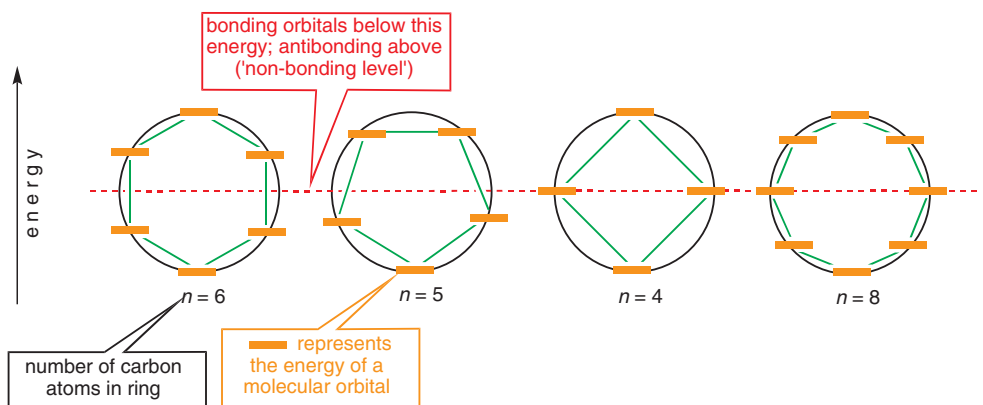
The difference between the amount of energy we expect to get from benzene on hydrogenation (360 kJ mol^{-1}) and what is observed (208 kJ mol^{-1}) is about 150 kJ mol^{-1} . This represents a crude measure of just how extra stable benzene really is relative to what it would be like with three localized double bonds. In order to understand the origin of this stabilization, we must look at the molecular orbitals. We can think of the π molecular orbitals of benzene as resulting from the combination of the six p orbitals in a ring and, as with butadiene, each successively higher energy orbital contains one more node. This is what we get for benzene:



The molecular orbital lowest in energy, ψ_1 , has no nodes, with all the orbitals combining in phase. The next lowest molecular orbital will have one nodal plane, which can be arranged in two ways depending on whether or not the nodal plane passes through a bond or an atom. It turns out that these two different molecular orbitals both have exactly the same energy, that is, they are degenerate, and we call them both ψ_2 . There are likewise two ways of arranging two nodal planes and again there are two degenerate molecular orbitals ψ_3 . The final molecular orbital ψ_4 will have three nodal planes, which must mean all the p orbitals combining out of phase. Six electrons slot neatly into the three lowest energy bonding orbitals.

The π molecular orbitals of other conjugated cyclic hydrocarbons

Notice that the layout of the energy levels in benzene is a regular hexagon with its apex pointing downwards. It turns out that the energy level diagram for the molecular orbitals resulting from the combination of *any* regular cyclic arrangement of p orbitals can be deduced from the appropriately sided polygon with an apex pointing downwards. The horizontal diameter (the red line) represents the energy of a carbon p orbital and any energy levels on this line represent non-bonding molecular orbitals. All molecular orbitals with energies below this line are bonding; all those above are antibonding.



Aromaticity of cyclic polyenes

It's worth noting a few points about these energy level diagrams:

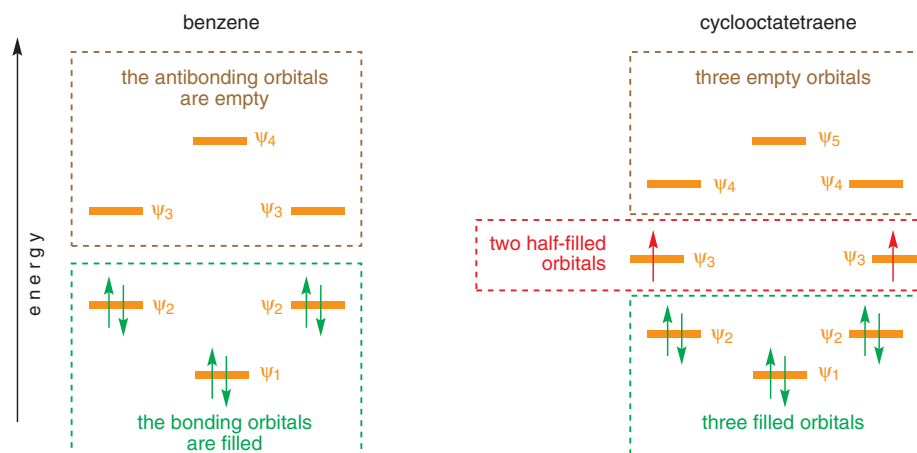
- The method predicts the energy levels for the molecular orbitals of planar, cyclic arrangements of identical atoms (usually all C) only.
- There is always one single molecular orbital lower in energy than all the others. This is because there is always one molecular orbital where all the p orbitals combine in phase.
- If there is an even number of atoms, there is also a single molecular orbital highest in energy; otherwise there will be a pair of degenerate molecular orbitals highest in energy.
- All the molecular orbitals come in degenerate pairs except the one lowest in energy and (for even-numbered systems) the one highest in energy.

Molecular orbitals and aromaticity

Now we can begin to put all the pieces together and make sense of what we know so far. We'll compare the way that the electrons fit into the energy level diagrams for benzene and planar cyclooctatetraene. We are not concerned with the actual shapes of the molecular orbitals involved, just their energies.

Benzene has six π electrons, which means that all its three bonding molecular orbitals are fully occupied, giving what we can call a 'closed shell' structure. Cyclooctatetraene's eight electrons, on the other hand, do not fit so neatly into its orbitals. Six of these fill up the bonding molecular orbitals but there are two electrons left. These must go into the degenerate pair of non-bonding orbitals. Hund's rule (Chapter 4) would suggest one in each. Planar cyclooctatetraene would not have the closed shell structure that benzene has—to get one it must either lose or gain two electrons. This is exactly what we have already seen—both the dianion and dication from cyclooctatetraene are planar, allowing delocalization all over the ring, whereas neutral cyclooctatetraene avoids the unfavourable arrangement of electrons shown below by adopting a tub shape with localized bonds.

■ You can draw an analogy here with the stability of 'closed shell' electronic arrangements in atoms.



Hückel's rule tells us if compounds are aromatic

As we pointed out on the previous page, all the cyclic conjugated hydrocarbons have a single lowest energy molecular orbital, and then a stack of degenerate pairs of orbitals of increasing energy. Since the single low energy orbital holds two electrons, and then the successive degenerate pairs four each, a 'closed shell' arrangement in which all the orbitals below a certain level are filled will always contain $(4n + 2)$ electrons (where n is an integer—0, 1, 2, etc.—corresponding to the number of degenerate orbital pairs). This is the basis of Hückel's rule.

● Hückel's rule

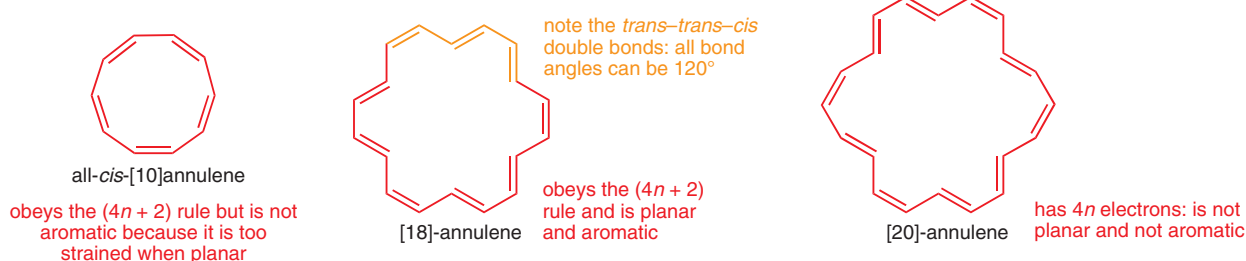
Planar, fully conjugated, monocyclic systems with $(4n + 2)$ π electrons have a closed shell of electrons all in bonding orbitals and are exceptionally stable. Such systems are said to be aromatic.

Analogous systems with $4n$ π electrons are described as anti-aromatic.

The next $(4n + 2)$ number after six is ten so we might expect this cyclic alkene, [10]annulene, to be aromatic. But if a compound with five *cis* double bonds were planar, each internal angle would be 144° . Since a normal double bond has bond angles of 120° , this would be far from ideal. This compound can be made but it does *not* adopt a planar conformation and therefore is not aromatic even though it has ten π electrons.

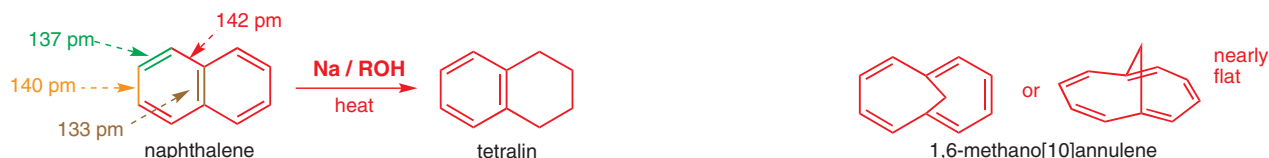
■ This is not a strict definition of aromaticity: it is actually very difficult to define aromaticity precisely, but all aromatic systems obey Hückel's $(4n + 2)$ rule.

Annulenes are compounds with alternating double and single bonds. The number in brackets tells us how many carbon atoms there are in the ring. Using this nomenclature, you could call benzene [6]annulene and cyclooctatetraene [8]annulene—but don't.

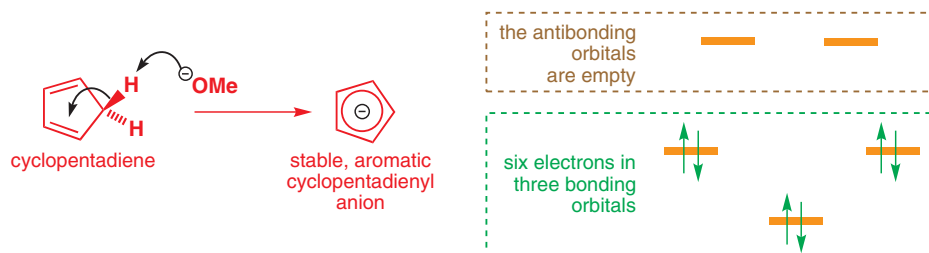


By contrast, [18]annulene, which is also a $(4n + 2)$ π electron system ($n = 4$), does adopt a planar conformation and *is* aromatic. The *trans-trans-cis* double bond arrangement allows all bond angles to be 120° . [20]Annulene presumably could become planar (it isn't quite) but since it is a $4n$ π electron system rather than a $4n + 2$ system, it is not aromatic and the structure shows localized single and double bonds.

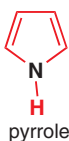
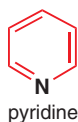
When the conjugated systems are not monocyclic, the situation becomes a little less clear. Naphthalene, for example, has ten electrons but you can also think of it as two fused benzene rings. From its chemistry, it is very clear that naphthalene has aromatic character (it does substitution reactions) but is less aromatic than benzene itself. For example, naphthalene can easily be reduced to tetralin (1,2,3,4-tetrahydronaphthalene), which still contains a benzene ring. Also, in contrast to benzene, all the bond lengths in naphthalene are not the same. 1,6-Methano[10]annulene is rather like naphthalene but with the middle bond replaced by a methylene bridging group. This compound is almost flat and shows aromatic character.



Hückel's rule helps us predict and understand the aromatic stability of numerous other systems. Cyclopentadiene, for example, has two conjugated double bonds but the conjugated system is not cyclic since there is an sp^3 carbon in the ring. However, this compound is relatively easy to deprotonate to give a very stable anion in which all the bond lengths are the same.



■ Not only are most aromatic systems heterocyclic, but more than 50% of *all organic compounds* contain an aromatic heterocycle.



Each of the double bonds contributes two electrons and the negative charge (which must be in a p orbital to complete the conjugation) contributes a further two, making six altogether. The energy level diagram shows that six π electrons completely fill the bonding molecular orbitals, thereby giving a stable aromatic structure.

Heterocyclic aromatic compounds

So far all the aromatic compounds you have seen have been hydrocarbons. However, most aromatic systems are heterocyclic—that is, they contain atoms other than just carbon and hydrogen. (In fact the majority of *all organic compounds* are aromatic heterocycles!) A simple example is pyridine, in which a nitrogen replaces one of the CH groups of benzene. The ring still has three double bonds and thus six π electrons.

Consider the structure shown on the left, pyrrole. This is also aromatic but it's not enough just to use the electrons in the double bonds: in pyrrole the nitrogen's lone pair contributes to the six π electrons needed for the system to be aromatic. Aromatic chemistry makes several more appearances in this book: in Chapter 21 we shall look at the chemistry of benzene and in Chapters 30 and 31 we shall discuss heterocyclic aromatic compounds in much more detail.

Further reading

Molecular Orbitals and Organic Chemical Reactions: Student Edition by Ian Fleming, Wiley, Chichester, 2009, gives an excellent account of delocalization.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>